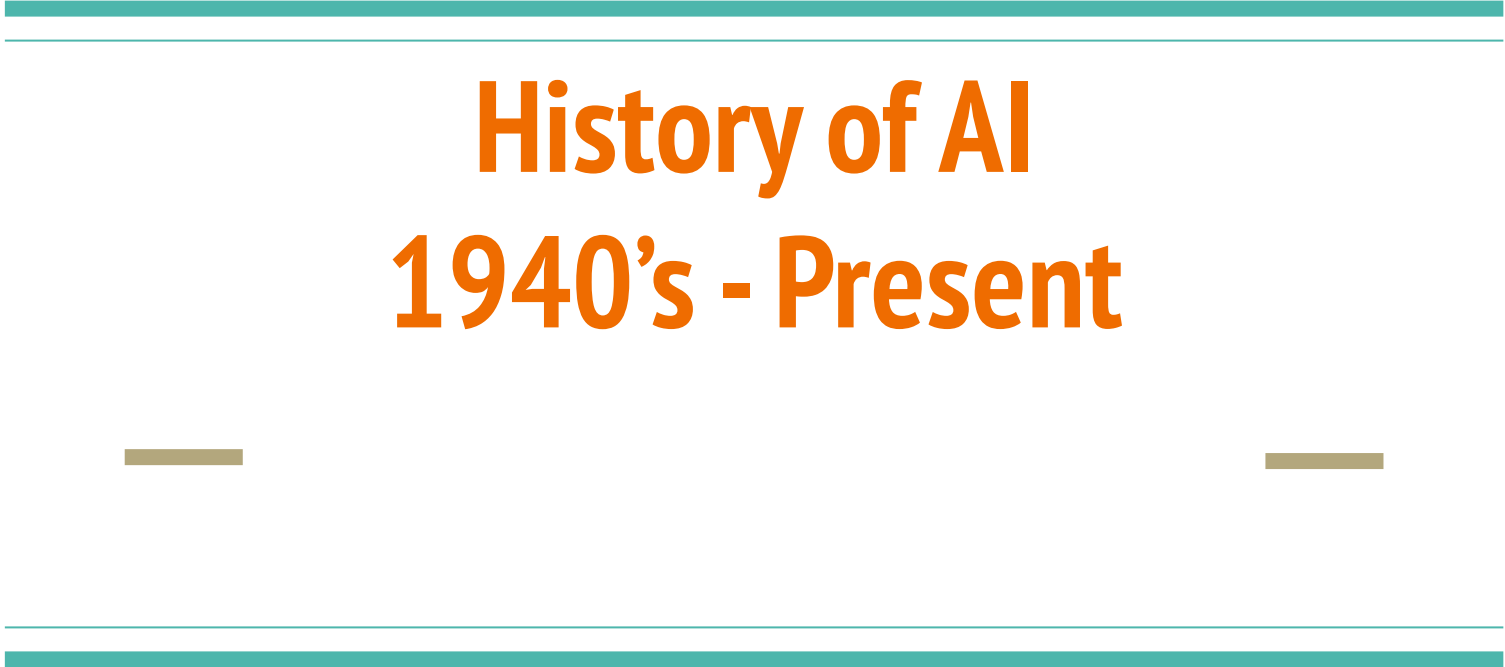
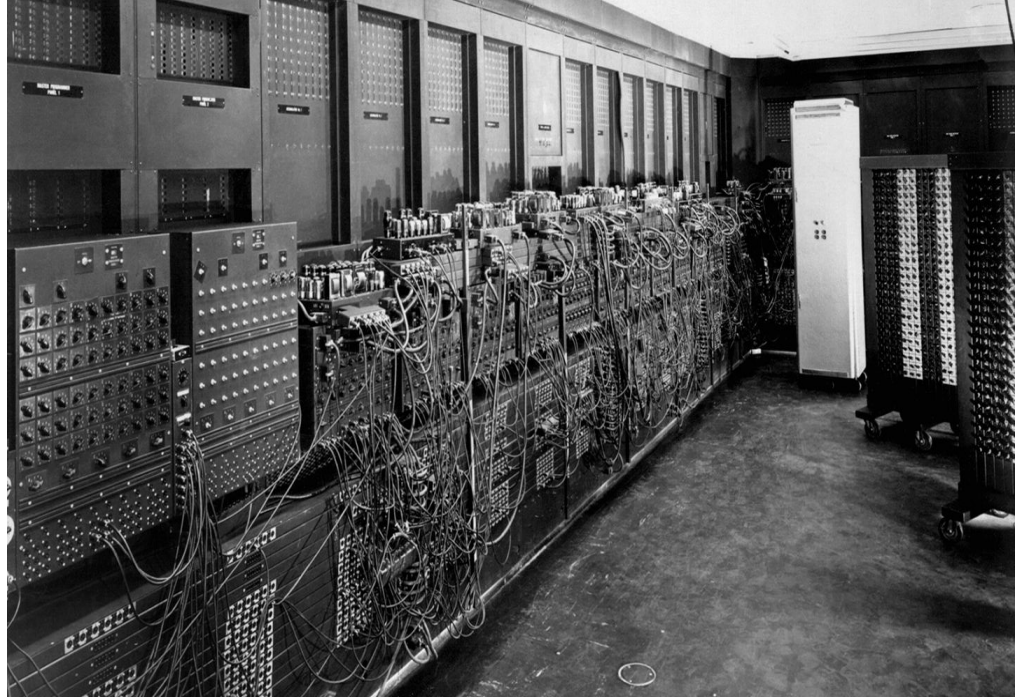

History of AI

1940's - Present

The slide features a light blue background with a central white rectangular area. This area is framed by two horizontal teal lines at the top and bottom. Inside the white area, there are two short, horizontal olive-green dashes, one on the left and one on the right, positioned below the main title.

Early Foundations

- **1943:** Warren McCulloch and Walter Pitts develop a mathematical model for neural networks.
- **1946:** The ENIAC (Electronic Numerical Integrator and Computer) is completed, marking a significant step in the development of computing technology.



1950: The Question

- **1950:** Alan Turing publishes "Computing Machinery and Intelligence," introducing the concept of the Turing Test to assess a machine's ability to exhibit intelligent behavior.

Turing asks the question:

“ Can machines think?”



How AI got its Name!

The name Artificial Intelligence is credited to John McCarthy who, along with Marvin Minsky, Nathaniel Rochester and Claude Shannon, organized the Dartmouth Conference in 1956.

Dartmouth Project Proposal: Proposal of a 2 month 10 man study of Artificial Intelligence be carried out at Dartmouth College in Hanover, New Hampshire.. “Two month, ten-man study of AI.. on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it”.



John McCarthy



Marvin Minsky

Dartmouth Conference: The Organizers

- John McCarthy (1927-2011)
 - an Assistant Professor at Dartmouth.
 - Designed the LISP programming language that was very popular with AI researchers.
 - Also did work in Logic and Commonsense Reasoning.
- Marvin Minsky (1927-2016)
 - a Harvard Junior Fellow went on to become one of the most influential figures in AI.
 - With McCarthy he co-founded the MIT AI Lab.
 - Known for his ideas on Frames. Wrote a book “Society of the Mind” and more recently “The Emotion Machine”.
- Nathaniel Rochester (1919-2001)
 - a young engineer at IBM.
 - He designed the IBM 701 and wrote the first assembler.
 - He supervised Arthur Samuel writing the checkers playing program.
 - It is said that the marketing people at IBM reported that people were frightened of “electronic brains” resulting in IBM stopping work on AI.
- Claude Shannon (1916-2001)
 - a mathematician at Bell Labs was already known for his information theory.
 - Had hired McCarthy and Minsky in 1952 for the summer when they were graduate students.

Dartmouth Conference: The Show Stealers

- Herbert Simon and Alan Newell were “two vaguely known persons” working at Carnegie Institute of Technology (now Carnegie Mellon University), who were also invited to the Dartmouth Conference “almost as an afterthought”.
- Simon was primarily known for his work in economics and organizational behavior, having laid the groundwork for the concept of bounded rationality.
- Newell was a computer scientist with a keen interest in cognitive psychology and human problem-solving.
- In collaboration with Simon, Newell worked on the General Problem Solver (GPS), an early AI program designed to imitate human problem-solving skills across a variety of domains.
- They had already developed program called the Logic Theorist, considered one of the first AI programs, which was capable of proving mathematical theorems.



Samuel's Checkers Program

- Arthur Samuel (1901-1990) was one of the attendees in Dartmouth.
- He began working on the checkers program in the early 1950s, precisely 1952 on IBM's 701.
- The goal was to create a program that could play checkers at a high level, ultimately improving its performance through learning from experience.



ELIZA - the Chatbot Psychotherapist

- Created by Joseph Weizenbaum at the MIT Artificial Intelligence Laboratory between 1964 and 1966.
- ELIZA was designed to simulate conversation with a psychotherapist, using pattern matching and substitution methodology to create the illusion of understanding.
- The program demonstrated how computers could be made to communicate with humans in natural language, laying the groundwork for future developments in conversational agents and natural language processing (NLP).

Welcome to

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EEEEEE	LLLLLL	IIII	ZZZZZZ	AA AA

Eliza is a mock Rogerian psychotherapist.

The original program was described by Joseph Weizenbaum in 1966.

This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?

YOU: Men are all alike.

ELIZA: What is the connection, do you suppose ?

YOU: They're always bugging us about something or other.

ELIZA: Can you think of a specific example ?

YOU: Well, my boyfriend made me come here.

ELIZA: Is it important to you that your boyfriend made you come here ?

YOU: He says I'm depressed much of the time.

ELIZA: I am sorry to hear that you are depressed.

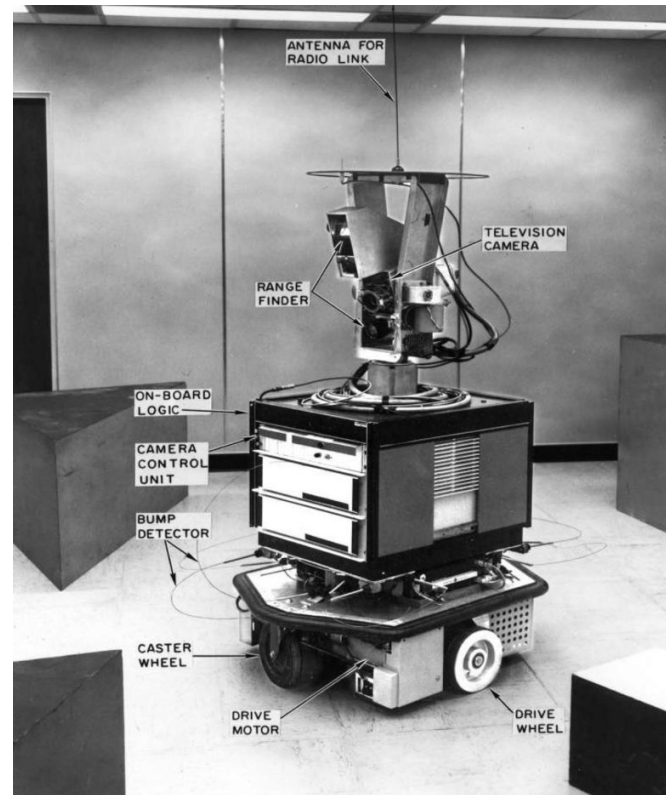
YOU: It's true. I am unhappy.

ELIZA: Can you explain what made you unhappy ?

YOU:

Shakey - the general purpose mobile robot

- Shakey the Robot was a pioneering project in robotics and artificial intelligence developed by the Stanford Research Institute (SRI) from 1966 to 1972.
- Shakey was the first general-purpose mobile robot capable of reasoning about its actions and operating autonomously in a structured environment.
- This project significantly advanced the fields of robotics, AI, and computer vision.
- Shakey could navigate through rooms, avoid obstacles, and move objects to specific locations. It combined data from its sensors to map its environment and plan paths.



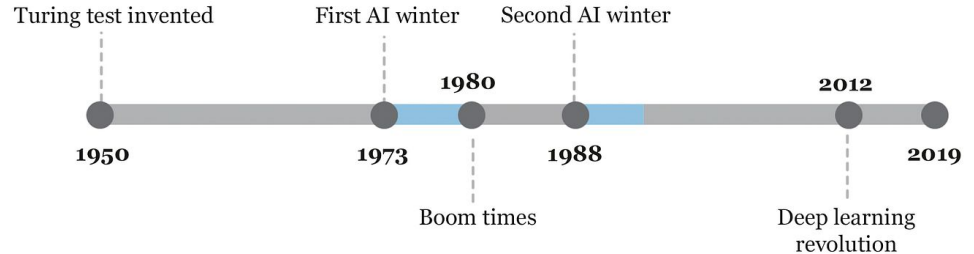
During 1956 - 1966

- 1950: Turing Test for Machine Intelligence
- 1956: AI born at Dartmouth College Workshop
- 1964: ELIZA - the Chatbot Psychotherapist
- 1966: Shakey - the general purpose mobile robot

The First AI Winter: 1974 - 1980

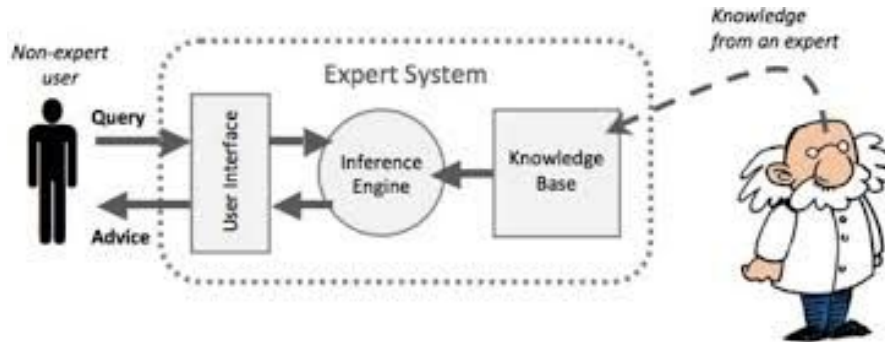
Despite initial enthusiasm, progress in AI research slows due to limitations in computational power and overly optimistic predictions, leading to reduced funding and interest. This period is known as the "AI Winter."

- Failure of Machine Translation
- Negative results in Neural Nets
- Poor speech understanding



Renewed Interest and Expert Systems

AI research experiences a resurgence with the development of expert systems, which are computer programs designed to mimic the decision-making abilities of human experts. Companies like IBM and others invest heavily in AI research.



The Second AI Winter: Late 1980s-1990s

The limitations of expert systems and other AI technologies lead to another period of reduced funding and interest, marking the second AI Winter.

- Decline of LISP
- Decline of specialized hardware for expert systems

Lasting effects:

- Economist'07: "AI is associated with systems that have all too often failed to live up to their promises"
- Pittsburgh BT06: "some believe the word "robotics" actually carry stigma that hurts a company's chances at funding"

Attendance at large conferences (1984-2018)

Source: Conference provided data

